Band-1 Hadamard Order 4916 — Miyamoto (Paley-II seed H_1228); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_1228/C_1230/H_4916 PASS; 4916 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4868_Miyamoto`; `2026-09-15_Band1_Hadamard_4084_Miyamoto` (Paley-II seed pattern).
Lab. `/workspace/experiments/band1/hadamard_4916_miyamoto/`.
Cutoff. 2026-09-15T17:59:58Z (~1:59PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1228 and 4916 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{1228} ($q = 613$). Paley-I direct for H_{1228} is **SKIP** ($q = 1227 = 3 \cdot 409$ not prime). Double Paley-I half = 614 is **FAIL** ($q = 613 \equiv 1 \pmod 4$; Paley-I needs $\equiv 3$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4916 is a **Miyamoto-only bonus** (NOT in SPK): Paley-I FAIL ($q = 4915 = 5 \cdot 983$), Paley-II FAIL ($q = 2457 = 3^3 \cdot 7 \cdot 13$ not prime power), Kronecker FAIL (4×1229). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4868 on box. This increment ships **order 4916** ($4916 = 4 \cdot 1229$, 1229 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Paley-II seed** H_{1228} ($q = 613$). Symmetric Paley conference C_{1230} over prime 1229; $H_{4916} = \text{Miyamoto}(H_{1228}, C_{1230})$. Unit tests **55/55 PASS**. SPK next gap remains **2052** (OPEN). Order 4916 is **Miyamoto-only** (NOT in SPK). Verified H_{1228} enters SPK as a Paley-II seed. The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4916; Miyamoto; Paley type II; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{1228} constructed and verified (Paley type II, $q = 613$). Paley-I direct **SKIP** ($q = 1227 = 3 \cdot 409$). Double Paley-I **FAIL** ($q = 613 \equiv 1 \pmod 4$). 2. C_{1230} constructed and verified (Paley conference, $q = 1229$): $CC^\top = 1229I$, zero diagonal. 3. H_{4916} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. SPK: $4916 \notin$ SPK (Miyamoto-only bonus). Paley-I / Paley-II / Kronecker paths for 4916 **rejected**. Next constructive gap remains **2052** (OPEN). Verified H_{1228} enters SPK as Paley-II seed. Prior 608/1216 (Double Paley-I) and 4804 (Paley-II 7^4) remain in SPK. Prior 4868 remains Miyamoto-only (excluded from SPK seeds).

2. Arithmetic and eligibility

Quantity	Value
Target order	$4916 = 4 \cdot 1229$
$q = 1229$	prime, $1229 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1228}
Needed conference	order $1230 = q + 1$
H_{1228} path (used)	Paley-II: $2(q+1) = 1228 \Rightarrow q = 613$ prime $\equiv 1 \pmod 4$
H_{1228} Paley-I direct	**SKIP**: $q = 1227 = 3 \cdot 409$ not prime
H_{1228} Double Paley-I	**FAIL**: half $= 614$, $q = 613 \equiv 1 \pmod 4$ (needs $\equiv 3$)
Conference path	Symmetric Paley over prime 1229
Paley-I for 4916	**REJECTED**: $q' = 4915 = 5 \cdot 983$ not prime
Paley-II for 4916	**REJECTED**: $q' = 2457 = 3^3 \cdot 7 \cdot 13$ not prime power

Quantity	Value
Kronecker for 4916	**REJECTED**: only 4×1229; $1229 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{1228}	Paley type II, $q = 613$	0	1.0414	**PASS**
H_{1228} Paley-I direct	$q = 1227 = 3 \cdot 409$	—	—	**SKIP** (not prime)
H_{1228} Double Paley-I	half $= 614$, $q = 613 \equiv 1$	—	—	**FAIL** (needs $\equiv 3$)
C_{1230}	Paley conference, $q = 1229$	0	1.5646	**PASS**
H_{4916}	Miyamoto(H_{1228}, C_{1230})	0	51.9555	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 56.1622 s. Compressed ± 1 matrices saved as `H1228_pm1.npz`, `C1230_pm1.npz`, `H4916_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 ∪ on-box prior classical/SPK ships (through 4868/4804/4036/3908/… + **1020/1116** Paley-II + **1032/1092/1152** Paley-I + **524/1048/548/1096/564/1128/600/1200/608/1216** Double Paley-I + **4804** Paley-II prime power 7^4; **exclude** Miyamoto-only 4868/4612/4516/4468/4388/4372/4196/4 ∪ verified H_{1228} (Paley-II) ∪ Sylvester ∪ Paley I/II (incl. prime powers), closed under Kronecker (limit 5000). Order 4916 is **Miyamoto-only** (not folded into SPK).

Quantity	Value
Known count	1047
4916 in SPK	**False** (Miyamoto-only bonus)
1228 in SPK	**True** (Paley-II seed)
1216 in SPK	**True** (prior Double Paley-I)
608 in SPK	**True** (prior Paley-I)
4804 in SPK	**True** (Paley-II prime power 7^4)
4868 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

Quantity	Value

5. Gap 2052 (method triage / non-finding ONLY)

Paley-I, Paley-II, Kronecker, and Miyamoto-with-Paley-conference paths for order 2052 all **FAIL** on-box arithmetic. This is a **non-finding**, not a counterexample. Gap **2052 remains OPEN**. Conjecture **OPEN**.

6. Non-claims

- No full Hadamard conjecture claim.
- `clay_closed = false` forever in this ship.
- GATE-04 collection **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / finished-ToE claims.
- Order 4916 is a **Miyamoto-only bonus** (NOT in SPK).
- Gap 2052 is **not** filled by this ship.

7. Reproduce

```
cd /workspace/experiments/band1/hadamard_4916_miyamoto
python3 verify_hadamard_4916.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `H1228_pm1.npz`, `C1230_pm1.npz`, `H4916_pm1.npz`, this paper.

8. Next optional

After $q = 1229$: **q=1237→n=4948** PaleyII($q = 617$), or **q=1249→n=4996** Kronecker(4×312), or method-attack SPK gap **2052**, or Track B privacy / Object 1 meter await.